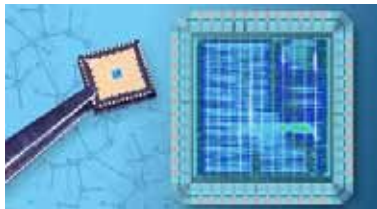


Truly Innovative **TECH**

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Tech*

SNN chip to process radar signals

Artificial neural networks (ANNs) are the backbone of artificial intelligence (AI) and machine learning (ML). However, ANNs consume too much power and data travel is a time-consuming procedure from the sensor device to the AI inference algorithm for decision making. To overcome this challenge, Imec, an



organisation doing R&D of nanoelectronics and digital technologies, has developed a chip that makes use of spiking neural networks (SNNs). Unlike ANNs, SNNs fire electrical pulses sparsely over time, and only when the sensory input changes. As such, energy consumption can significantly be reduced. Besides ECG and speech processing applications, the chip is a smart solution for implementing a low-power, anti-collision system for drones and autonomous cars.

Imec
www.imec-int.com

Automotive image sensor for low light

Enhanced with near-infrared (NIR) technology, the 2.5-megapixel OX03A2S image sensor is ideal for exterior imaging applications that operate in low to no ambient light conditions within two metres of the vehicle. It has a novel silicon semiconductor architecture, which helps to achieve automotive quantum efficiency of 40 per cent at 940nm NIR wavelength. This enables detection and recognition of objects that other image sensors would miss under extremely

low lighting conditions, enabling more capable safety systems. Automotive cameras using the OX03A2S require less illumination, thus reducing material cost and overall power consumption.



With the ability to display high quality, NIR-enhanced, RGB, and machine vision images, automotive designers can efficiently utilise the sensor for both day and night environments.

Omnivision Technologies
www.ovt.com

Embedded sensor for smart tyres

To facilitate tyre-pressure monitoring for commercial vehicles, including heavy-duty and passenger cars, and enable future generations of smart tyres, the



MLX91805 has been specially designed to be embedded in the tyre, unlike current rim-mounted TPMS sensors. The sensor integrates an 800g XZ-axis accelerometer that can sense up to 10,000 samples per second. It also integrates a high-accuracy pressure sensor as well as temperature and voltage sensors. Integrating a low-power 16-bit microcontroller and 315/433MHz RF transmitter that draws just 5mA, and with an extremely low 90nA sleep current, the MLX91805 can operate from a tiny battery for the life of the tyre. In addition to built-in diagnostics

and fault detection, a math coprocessor allows fast and energy-efficient digital signal processing.

Melexis
www.melexis.com

Fuel gauge IC with monitoring feature

The new MAX17320 high-accuracy fuel-gauge and protection IC extends run-time on multi-cell battery-powered products and also keeps in check the self-discharge hazards. For up to 2 to 4 series Lithium-ion (Li+) cells (2S-4S), the device is part of an IC family equipped with ModelGauge m5 EZ



algorithm that delivers 40% more accurate state-of-charge (SOC) readings, eliminating the need for battery characterisation for most common Li+ cells. This fuel gauge also offers 85% lower quiescent current and features SHA-256 authentication for safeguarding systems from counterfeit batteries. Until now there were no solutions for detecting self-discharge that develops during normal usage of the battery. But with the MAX17320, which is suitable for handheld devices with 2S-4S Li+ batteries, applications consisting of computing, IoT, power tool, consumer, healthcare, and mobile device categories can be met with high customer safety standards.

Maxim Integrated
www.maximintegrated.com

Portable LiDAR with powerful features

This all-in-one solid-state LiDAR comes with powerful control features, which makes the system more portable than